1 The Real and Complex Number Systems

1.1 The Rational Numbers

Theorem 1.1.1 (Irrationality of $\sqrt{2}$): For all $x \in \mathbb{Q}$, $x \neq \sqrt{2}$

Proof,

Assume otherwise. Thus, there exists a $x \in \mathbb{Q}$ such that $x = \sqrt{2}$.

Now, since $x \in \mathbb{Q}$, it follows that:

$$x = \frac{a}{b}$$

where, $a \in \mathbb{Z}, b \in \mathbb{Z} \setminus \{0\}$. The trick here is to add the further restriction that $\gcd(a, b) = 1$.

Whilst I won't prove it, it should be sufficient understandable why coprimality can be assumed.

Now, square both sides:

$$x^2 = \frac{a^2}{b^2} = 2 \implies a^2 = 2b^2$$

Thus, a is even hence $a = 2k$.

$$2b^2 = 4k^2 \implies b^2 = 2k^2$$

Thus, b is also even.

However, we had assume coprimality but we just showed $\gcd(a, b) = 2$.

This is a contradiction, so x can't exist.

□

Theorem 1.1.2 ($\mathbb{Q}$ can have sets bounded above without a least upper bound): Consider:

$$A = \{x \in \mathbb{Q} : x^2 < 2\} = \left(-\infty, \sqrt{2}\right) \cap \mathbb{Q}$$

Then, A is bounded from above but A doesn't have a least upper bound.

Definition 1.1.1 (Bounded Above): Let $X \subseteq \mathbb{Q}$, then X is said to be bounded above if it has an upper bound i.e., a number $u \in \mathbb{Q}$ such that:

$$\forall x \in X, x \leq u$$

Definition 1.1.2 (Least Upper Bound): Let $X \subseteq \mathbb{Q}$, then X is said to be have a least upper bound if there exists a number $s \in \mathbb{Q}$ satisfying these properties:
1. x is an upper bound
2. If $y \in \mathbb{Q}$ is an upper bound of X, then $x \leq y$

Proof,

For contradiction, assume that A has a least upper bound s. By trichotomy,

$$s^2 < 2 \quad s^2 = 2 \quad s^2 > 2$$

(Trichotomy is valid for $\mathbb{Q}$ also)

However, since $s^2 = 2 \implies s = \sqrt{2}$ this implies that $s^2 \neq 2$. Thus, we only have two cases to deal with.

For the first case $s^2 < 2$, we show that s^2 is not an upper bound.

Consider,

$$\left(s+\frac{1}{n}\right)^2 = s^2 + \frac{2s}{n} + \frac{1}{n^2}$$

Let $\varepsilon = 2 - s^2 > 0$, thus,

$$\left(s+\frac{1}{n}\right)^2 - 2 = -\varepsilon + \frac{2s}{n} + \frac{1}{n^2}$$

By Archimedian property of $\mathbb{Q}$,

$$\frac{1}{n_1} < \frac{\varepsilon}{6s}$$

This is valid since $s \geq 1$ as $1 \in A$.

Similarly, we can have:

$$\frac{1}{n_2^2} < \frac{\varepsilon}{3}$$

Let, $n = \max\{n_1, n_2\}$. Then,

$$\left(s+\frac{1}{n}\right)^2 - 2 = -\varepsilon + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < 0$$

Thus, we're done since $\left(s+\frac{1}{n}\right)^2 < 2 \implies s + \frac{1}{n} \in A \implies s$ can't be an upper bound since a number greater than it is within A.

For the next $s^2 > 2$, we show that s is not the least upper bound.

This follows because,

$$\left(s-\frac{1}{n}\right)^2 - 2 = \varepsilon - \frac{2s}{n} + \frac{1}{n^2} > \varepsilon - \frac{2s}{n}$$

For the last part, simply recall that by Archimedian property, we get:

$$\frac{2s}{n} < \frac{\varepsilon}{2} \implies -\frac{2s}{n} > \frac{\varepsilon}{2}$$

This follows because of the fact that $1 \in A \implies s \geq 1 > 0$. Thus, we've;

$$\left(s - \frac{1}{n}\right)^2 - 2 > \varepsilon - \frac{\varepsilon}{2} = \frac{\varepsilon}{2} > 0$$

Hence, we have another contradiction.

With all options from trichotomy invalidated, it follows that s can't exist in $\mathbb{Q}$.

□

We shall prove some interesting results. The first is the Archimedian property which was so important in the preceding proof.

□

Theorem 1.1.3 (Archimedian Property of $\mathbb{Q}$): Let $q \in \mathbb{Q}, q \geq 0$, then there exists a $n \in \mathbb{N}$ such that:

$$q \leq n$$

Proof,

Before we prove it, we need a lemma.

Lemma 1.1.4: $\mathbb{N}$ is unbounded from above.

Being unbounded above from above means it's not bounded from above. Thus, written formally:

$$\forall n \in \mathbb{N}, \exists m \in \mathbb{N} : n \leq m$$

We shall prove this result by contradiction.

So assuming the negation:

$$\exists n \in \mathbb{N} : \forall m \in \mathbb{N}, n > m$$

Since, $n \in \mathbb{N} \implies n + 1 \in \mathbb{N}$ (inductive property of $\mathbb{N}$).

But,

$$n + 1 > n$$

This contradicts the assumption, so $\mathbb{N}$ must be unbounded from above.

Now, $q \in \mathbb{Q}^+$, so:

$$q = \frac{a}{b}$$

Assume that a, b have no common factors except 1. This forces $a, b \in \mathbb{N}$.

Now,

$$qb = a \in \mathbb{N}$$

Due to the fact that $\mathbb{N}$ is unbounded, there exists a n such that:

$$qb = a < n < bn$$

Hence,

$$q < n$$

□

Archimedian Property is intimately connected with limits. In fact, this "limit" is how we proved that there bounded sets in $\mathbb{Q}$ without a least upper bound. Since, it's with Archimedian property that we can confidently say that $\frac{1}{n}, \frac{1}{n^2}$ tends to vanish (i.e., become zero) as n becomes larger.

Note that we will make the last sentence more rigorous.

□

An interesting property of mathematical structures, especially ones that are interest to analysis, is that of density.

I won't be defining it formally here (since we can define in generality with only a bit more tools). However, informally I will.

□

Theorem 1.1.5 (Density of $\mathbb{Q}$): $\mathbb{Q}$ is dense in $\mathbb{Q}$.

Proof,

So, what does density mean here? It basically means that if I have two $x, y \in \mathbb{Q}$ such that $x \neq y$, I can find another rational number $r \in \mathbb{Q}$ between x, y i.e.,

$$\forall x, y \in \mathbb{Q}, x < y \implies \exists r \in \mathbb{Q} : x < r < y$$

Note that just proving one implies an infinitely many choices of r because we basically recurse over and over again between r, y or r, x. This is annoying to show rigorously (it requires induction), but should appeal to intuition.

Now, the easy choice of r is simply the usual arithmetic mean:

$$r = \frac{x + y}{2}$$

Now

$$r - x = \frac{x + y}{2} - x = \frac{y - x}{2} > 0$$

Since, $y - x > 0$.

Similarly,

$$y - r = y - \frac{x + y}{2} = \frac{y - x}{2} > 0$$

and we're done.

□

Now I want you to think really hard about what this result brings.

We have proven that $\mathbb{Q}$ has gaps. However, we have also shown that no matter how small of a gap is between two rational numbers, we can always find another (rational) number in between.

This fact is significant. It shows that there is some structural defect in $\mathbb{Q}$.

□

1.2 Ordered Sets

Definition 1.2.1 (Order): Let S be a set. An order on S is a relation, denoted by $<$, with the two following properties:

1. (Trichotomy) If $x \in S$, $y \in S$, the one and only one of the statements is true:

$$x < y \quad x = y \quad y < x$$

2. (Transitivity) if $x, y, z \in S, x < y, y < z \implies x < z$

□

Definition 1.2.2 (Ordered Set): An ordered set S is a set S which as an order defined on it.

□

Before we move forward, I wish to recall the definition of relation.

Definition 1.2.3 (Cartesian Product): Let A, B be sets. Then the Cartesian product is:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

With this,

Definition 1.2.4 (Relation): Let A be a set. Then a relation on A is a subset of $A \times A$

Relations are really general constructs. They lack much structure as such we really only care about special restricted relations (functions, equivalence relations, orderings, etc.).

□

With the concept of ordered set down, we can noe generally define bounds.

Definition 1.2.5 (Upper Bound): Let S be an ordered set. If $E \subseteq S$, then an upper bound of E is an element $b \in S$ such that:

$$e < b \forall e \in E$$

Definition 1.2.6 (Bounded Above): Let S be a ordered set. Then $E \subseteq S$ is said to be bounded from above if it has an upper bound.

□

We define lower bounds similarly.

□

Finally, we can define the least upper bound and the greatest lower bounds. Which are more commonly called supremum and infimum.

Definition 1.2.7 (Supremum): Let S be an ordered set, such that $E \subseteq S$. Then we say that E has a supremum if there exists $s \in S$ such that:
1. s is an upper bound of E.
2. If u is an upper bound of E, $s \leq u$

We denote,

$$s = \sup E$$

Remark: The notation $s \leq u$, means either $s < u$ or $s = u$

Definition 1.2.8 (Infimum): Let S be an ordered set, such that $E \subseteq S$. Then we say that E has an infimum if there exists $g \in S$ such that:
1. g is a lower bound
2. If l is a lower bound of E, then $l \leq g$

We denote,

$$g = \inf E$$

□

Definition 1.2.9 (Least Upper Bound Property): An ordered set S is said to have the least upper bound property if the following is true:

If $E \subseteq S$, $E \neq \varnothing$, and E is bounded from above, then E has a least upper bound.

□

Theorem 1.2.1 (Least Upper Bound Property equivalent to Greatest Upper bound Property): Let S be an ordered set with the least-upper-bound property, let $B \subseteq S$, $B \neq \varnothing$, B bounded below. Let L be the set of all lower bounds of B. Then,

$$\inf B = \sup L$$

Proof,

Why is this an "equivalence"? Well, the arguments really don't change much if we assumed the greatest-lower-bound property. It should be pretty evident what that property is (just reverse the bound in Definition 1.2.9).

Now, consider,

$$L = \{l \in S : l \text{ is a lower bound of } B\}$$

Because, B is nonempty, L is also nonempty. Moreover, $L \subseteq S$ and L is bounded from above by B.

By the least upper bound property, we have that:

$$\sup L$$

exists.

Now, $\sup L$ is a lower bound of B since:

$$\sup L \leq b$$

for all $b \in B$ because b is an upper bound of B.

Suppose l is a lower bound of B. by the definition of the supremum,

$$l \leq \sup L$$

Thus, $\sup L$ satisfies the definition for $\inf B$.

□

1.3 Fields

A Field is an algebraic structure (at this just point, just think of an algebraic structure as something that is a set + operations + rules of operations) that encodes the properties of what we usually consider "numbers."

Of course, this is not the case, since $\mathbb{N}$ is also considered to be numbers but doesn't define a field.

However, for analysis, it's a very important algebraic structure. And for us, it's the only algebraic structure we really care about.

Definition 1.3.1 (Field): A field is a set $\mathbb{F}$ with two operations called addition and axiom which satisfy the field axioms.

The field axioms are long and lengthy. So, here we list them all:

Le $\mathbb{F}$ be a field. Then $+$ is addition and $\cdot$ is multiplication. $a, b, c \in \mathbb{F}$.

Axiom of addition

1. (Closure) $a + b \in \mathbb{F}$
2. (Commutativity) $a + b = b + a$
3. (Associativity) $(a + b) + c = a + (b + c)$
4. (Existence of Identity) There exists $0 \in F$ such that: $a + 0 = a$

5. (Existence of Inverse) For all $a \in \mathbb{F}$, there exists an element a' such that $a + a' = 0$

Axioms for multiplication (For simplicity we denote $x \cdot y = xy$)

1. (Closure) $ab \in \mathbb{F}$
2. (Commutativity) $ab = ba$
3. (Associativity) $(ab)c = a(bc)$
4. (Existence of Identity) There exist $1 \in \mathbb{F} \neq 0$ such that $a \cdot 1 = a1 = a$ for all $a \in \mathbb{F}$.
5. (Existence of Inverse for non-zero element) If $a \neq 0$,then there exists $a' \in \mathbb{F}$ such that $aa' = 1$

   Remark: The 0 in multiplication is the additive identity. Moreover, there are algebraic structure which are simpler (rings and groups).

Finally, we have the axiom which connects multiplication and addition.

Axiom of Distributivity

1. (Distributivity) $a(b + c) = ab + ac$

□

Remark: Now, the most common field we know so far is $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ (the rational, real, and complex numbers respectively). There are more fields than this, of course. However, we don't particularly care for fields besides these.

□

Theorem 1.3.1 (Some properties of fields): Let $\mathbb{F}$ be a field, $x, y \in \mathbb{F}$. Then:

1. $x + y = x + z \implies y = z$
2. $x + y = x \implies y = 0$
3. $x + y = 0, y = -x$
4. $-(-x) = x$

Proof,

(1.) This is the cancellation law. This fact is true for any group.

$$x + y = x + z$$

We know that the additive inverse $-x$ exists. So,

$$-x + (x + y) = -x + (x + z) \Longrightarrow y = z$$

(2.) From $(1.)$,

$$x + y = x + 0 \Longrightarrow y = 0$$

(3.) From $(1.)$,

$$x + y = x - x \Longrightarrow y = -x$$

The importance of this result is that the additive inverse is unique.

(4.) This one requires a bit more work.

$$-(-x) + (-x) = (-x)(-1 + 1) = (-x)(0) = 0$$

Of course, we used distributivity in the second expression. For the final result, we did use that for any $x \in \mathbb{R}$:

$$x(0) = 0$$

To see why, observe:

$$x(0) = x(0 + 0) = x(0) + x(0) \Longrightarrow x(0) = 0$$

By (2.).

□

Theorem 1.3.2: The axioms of multiplication imply the following statements.

1. If $x \neq 0, xy = xz \implies y = z$
2. If $x \neq 0, xy = x \implies y = 1$
3. If $x \neq 0, xy = 1 \implies y = \frac{1}{x}$
4. If $x \neq 0 \implies \frac{1}{\frac{1}{x}} = x.$

Proof,

The key axiom here is the existence of multiplicative inverse for non-zero real numbers.

(1.) Here,

$$xy = xz$$

Since, the multiplicative inverse $\frac{1}{x}$ exists. So,

$$\frac{1}{x}(xy) = \frac{1}{x}(xz) \implies y = z$$

(2.) Here,

$$xy = x = x(1)$$

From (1.), it follows that $y = 1$.

(3.) Here,

$$xy = 1 = x\left(\frac{1}{x}\right)$$

By (1.) it follows that $y = \frac{1}{x}$.

The importance of this result is that it states that the multiplicative identity is unique.

(4.) The main result we neeed to show here is that the multiplicative inverse is non-zero.

I will argue by contradiction.

Suppose $x \neq 0$ but $\frac{1}{x} = 0$. Thus,

$$x\left(\frac{1}{x}\right) = 0 = 1$$

We know that $0 \neq 1$ by the multiplication axiom for Fields, this gives us a contradiction. Note that to get to zero, we used the result derived in (4.) of the previous theorem.

Therefore, $x \neq 0$ then its multiplicative is also non-zero.

With that out of the way, we can use (3.) to get:

$$x\left(\frac{1}{x}\right) = 1 \implies x = \frac{1}{\frac{1}{x}}$$

□

Theorem 1.3.3: The field axioms imply the following statements:

1. $0x = 0$
2. If $x, y \neq 0 \implies xy \neq 0$
3. $(-x)y = -(xy) = x(-y)$
4. $(-x)(-y) = xy$

Proof,

(1.) We use distributivity to get:

$$0x = (0 + 0)x \implies 0x = 0$$

By cancellation law of addition.

(2.) Suppose, there exists $x, y \in \mathbb{R}$ such that:

$$x, y \neq 0 \quad \text{but} \quad xy = 0$$

Then,

$$x^{-1}(xy) = y = 0$$

Hence, $y = 0$ which is a contradiction. Thus,

$$\forall x, y \in \mathbb{R}, x, y \neq 0 \implies xy \neq 0$$

(3.) Here, we have to prove for all $x, y \in \mathbb{R}$:

$$(-x)y = -(xy) = x(-y)$$

The proof here again needs to use distributivity alongside existence and uniqueness of the additive inverse:

$$xy + (-x)y = (x - x)y = x(y - y) = xy + x(-y) = 0$$

Thus,

$$(-x)y = x(-y) = -(xy)$$

The reason is that both $(-x)y, x(-y)$ be have like the additive inverse of xy. Thus, by the uniqueness of additive inverse,,

$$(-x)y = x(-y) = -(xy)$$

(4.) Finally,

$$(-x)(-y) = -(x(-y)) = -(-(xy)) = xy$$

which is a multi-fold application of (3.).

□

Definition 1.3.2 (Ordered Field): An ordered field is a field $\mathbb{F}$ which satisfies the following:

1. $\mathbb{F}$ is an ordered set
2. $\forall x, y, z \in \mathbb{F}, y < z \implies x + y < x + z$
3. $\forall x, y \in \mathbb{F}, x > 0, y > 0 \implies xy = 0.$

If $x > 0$, then x is positive and $x < 0$, then x is negative.

Remark: Recall that $\mathbb{F}$ being an ordered field implies the existence of the relation $<$ which satisfies trichotomy and transitivity.

So, unfurling the definition leads us that there are a total of four conditions required for a field to be considered ordered.

□

Theorem 1.3.4: The following statements are in every ordered field.

1. If $x > 0 \implies -x < 0$ and vice-versa.
2. If $x > 0, y < z$ then $xy < xz$
3. If $x < 0, y < z \implies xy > xz$
4. If $x \neq 0 \implies x^2 > 0$. In particular, $1 > 0$.
5. If $0 < x < y \implies 0 < \frac{1}{y} < \frac{1}{x}$.

Proof,

(1.) Here, we use the property that if $y < z$, then for any $x \in \mathbb{R}$:

$$y + x < z + x$$

Thus, if $0 < x$, we've:

$$0 - x < x - x \implies -x < 0$$

Similarly if $x < 0$, we've:

$$x - x < 0 - x \implies 0 < -x$$

The takeaway from this theorem is that the minus sign changes sign.

(2.) Let $x, y, z \in \mathbb{R}$ such that $x > 0, y < z$.

Now,

$$y < z \implies y - y < z - y$$

Thus,

$$z - y > 0$$

As $x > 0$, we've:

$$x(z-y) > 0 \Longrightarrow xz - xy > 0$$

Adding zy to both sides, we get:

$$xy < xz$$

(3.) For this, do the same operations like in (2.) until,

$$z - y > 0$$

Since, $x < 0 \Longrightarrow -x > 0$.

So,

$$-x(z-y) = -xz + yz > 0$$

Adding $-xz$ to both sides, we get:

$$xz < xy$$

(4.) This is an application of (3.)

If $x > 0$, remember that:

$$0(x) < x(x) \Longrightarrow 0 < x^2$$

If $x < 0$, remember that:

$$0(x) < x(x) \Longrightarrow 0 < x^2$$

(4.) Let $x, y \in \mathbb{R}$ such that:

$$0 < x < y$$

This statement is a bit of liar because there are actually two important statements here.

The first one is that,

$$\frac{1}{x} > 0$$

and that,

$$\frac{1}{x} > \frac{1}{y}$$

Let's prove it one by one.

$$x > 0$$

We know that $\frac{1}{x} \neq 0$, so by trichotomy,

$$\frac{1}{x} > 0 \text{ or } \frac{1}{x} < 0$$

If $\frac{1}{x} < 0 \implies x\left(\frac{1}{x}\right) < 0\left(\frac{1}{x}\right) \implies 1 < 0$.

Since $1^2 = 1 \implies 1 > 0$. This contradicts that fact, thus,

$$\frac{1}{x} > 0$$

With this done, we only need to prove that:

$$\frac{1}{y} < \frac{1}{x}$$

Now,

$$0 < x < y$$

Since, $\frac{1}{y} > 0$, it follows:

$$0 < \frac{x}{y} < 1$$

Similarly, $\frac{1}{x} > 0$, we also have:

$$0 < \frac{1}{y} < \frac{1}{x}$$

and so we're done. □

1.4 The Real Field

We won't prove it now. But eventually, we'll give two proofs of the following result. The first one, which will be given at the end of this section, will be the construction of $\mathbb{R}$ from $\mathbb{Q}$ using Dedekind cuts. The second method will given later.

For the mean time, we have this theorem:

Theorem 1.4.1 (Existence of $\mathbb{R}$): There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Remark: The second statement means that $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication of $\mathbb{R}$ when applied to $\mathbb{Q}$ cause $\mathbb{Q}$ to satisfy the field axioms.

□

Despite us proving the Archimedian property for $\mathbb{Q}$, we will need to once again reprove it for $\mathbb{R}$. Why? $\mathbb{R}$ adds new numbers, so we must make sure one of the numbers it added isn't an upper bound for $\mathbb{N}$.

Theorem 1.4.2 (Archimedian Property of $\mathbb{R}$): If $x, y \in \mathbb{R}$ and $x > 0$, then there exists a $n \in \mathbb{N}$ such that:

$$nx > y$$

Proof,

The proof here uses the least upper bound property.

First define,

$$z = \frac{y}{x}$$

This is valid as $x > 0$ which by Trichotomy means $x \neq 0$ thus the inverse exists.

Now that we've z, assume for contradiction that for all $n \in \mathbb{N}$, $z \geq n$. Thus, $\mathbb{N}$ has an upper bound in $\mathbb{R}$.

By the least upper bound property,

$$\sup \mathbb{N}$$

exists.

And by the arbitrary closeness criterion, there exists a $n \in \mathbb{N}$ such that:

$$n + 1 > \sup \mathbb{N}$$

Of course, this immediately creates a contradiction since $\mathbb{N}$ is inductive i.e., $n \in \mathbb{N} \implies n + 1 \in \mathbb{N}$. So,

$$n + 1 \leq \sup \mathbb{N} \quad \text{and} \quad \sup \mathbb{N} < n + 1$$

Thus, there exists a $n \in \mathbb{N}$ such that:

$$n > z$$

As $x > 0$, by the order property of $\mathbb{R}$, we get:

$$y < nx$$

□

Theorem 1.4.3 ($\mathbb{Q}$ is dense in $\mathbb{R}$): If $x, y \in \mathbb{R}$ and $x < y$, then there exists a $r \in \mathbb{Q}$ such that $x < p < y$